

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 3**Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
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Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A mass m on a spring of spring constant k oscillates with period T . If the mass is replaced by one of mass $2m$ (with the same spring), what is the new period? [2 marks]

- A. $T/\sqrt{2}$
- B. T
- C. $\sqrt{2}T$
- D. $2T$

2. A particle undergoes simple harmonic motion with amplitude A . At what displacement from equilibrium is the kinetic energy equal to three times the potential energy? [2 marks]

- A. $A/4$
- B. $A/\sqrt{3}$
- C. $A/2$
- D. $A/\sqrt{2}$

3. An underdamped oscillator has quality factor $Q = 50$. Approximately how many complete oscillation cycles does it take for the amplitude to fall to $1/e$ of its initial value? [2 marks]

- A. $\pi/Q \approx 0.063$ cycles
- B. $Q/(2\pi) \approx 8$ cycles
- C. $Q/\pi \approx 16$ cycles
- D. $2\pi Q \approx 314$ cycles

4. A driven damped oscillator has natural frequency ω_0 and damping constant γ . At what driving frequency ω is the phase lag δ between the driving force and the displacement exactly $\pi/2$? [2 marks]

- A. $\omega = \omega_0/2$
- B. $\omega = \omega_0$
- C. $\omega = \omega_0 \sqrt{1 - \gamma^2/(2\omega_0^2)}$
- D. $\omega = \sqrt{\omega_0^2 + \gamma^2}$

5. A transverse pulse travels along a taut string and arrives at an end that is rigidly fixed to a wall. Which of the following best describes what happens? [2 marks]

- A. The pulse is transmitted into the wall with no reflection.
- B. The pulse is reflected with no change in phase (upright).
- C. The pulse is reflected with a phase change of π (inverted).
- D. The pulse is absorbed and the string returns to rest.
- 6.** A string has a linear mass density $\mu = 0.04$ kg/m and is held under a tension $T = 100$ N. What is the impedance Z of the string? [2 marks]
- A. 0.8 kg/s
- B. 2.0 kg/s
- C. 4.0 kg/s
- D. 50 kg/s
- 7.** If the intensity of a sound wave is doubled, by how many decibels does the sound level increase? [2 marks]
- A. 1.5 dB
- B. 3.0 dB
- C. 6.0 dB
- D. 10 dB
- 8.** A source emitting sound at frequency $f = 500$ Hz moves toward a stationary observer at $v_s = 20$ m/s. At the same time, the observer moves toward the source at $v_o = 10$ m/s. Taking the speed of sound as $v = 340$ m/s, what is the frequency heard by the observer? [2 marks]
- A. 547 Hz
- B. 531 Hz
- C. 469 Hz
- D. 556 Hz
- 9.** A string of length L is fixed at both ends and vibrates in its 4th harmonic. How many nodes are present on the string (including the endpoints)? [2 marks]
- A. 3
- B. 4
- C. 5
- D. 8

10. Two identical simple pendulums of length L and mass m are connected by a light spring of spring constant k_c . Which of the following correctly describes the two normal modes of oscillation? [2 marks]

- A. Both pendulums swing in phase at frequency $\sqrt{g/L}/(2\pi)$; both swing in antiphase at frequency $\sqrt{g/L + 2k_c/m}/(2\pi)$.
- B. Both swing in phase at frequency $\sqrt{g/L + k_c/m}/(2\pi)$; both swing in antiphase at frequency $\sqrt{g/L + k_c/m}/(2\pi)$.
- C. Both swing in phase at frequency $\sqrt{g/L + 2k_c/m}/(2\pi)$; both swing in antiphase at frequency $\sqrt{g/L}/(2\pi)$.
- D. Both swing in phase at frequency $\sqrt{g/L}/(2\pi)$; both swing in antiphase at frequency $\sqrt{g/L + k_c/m}/(2\pi)$.

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. In Dirac notation, what type of mathematical object is $\langle a|b\rangle \langle c|d\rangle$? [2 marks]
- A. A ket
 - B. A bra
 - C. An operator
 - D. A scalar (complex number)
2. Which of the following represents a valid (normalised) quantum state? [2 marks]
- A. $|\psi\rangle = |1\rangle + |2\rangle$
 - B. $|\psi\rangle = \frac{1}{2} |1\rangle + \frac{1}{2} |2\rangle$
 - C. $|\psi\rangle = \frac{1}{\sqrt{2}} |1\rangle + \frac{i}{\sqrt{2}} |2\rangle$
 - D. $|\psi\rangle = \frac{1}{\sqrt{2}} |1\rangle + |2\rangle$
3. If $\{|n\rangle\}$ is a complete orthonormal basis, what does $\sum_n |n\rangle\langle n|$ equal? [2 marks]
- A. 0
 - B. The Hamiltonian $\hat{H}$
 - C. The identity operator $\hat{I}$
 - D. $\sum_n \langle n|n\rangle$
4. Given two operators $\hat{A}$ and $\hat{B}$, which of the following equals $(\hat{A}\hat{B})^\dagger$? [2 marks]
- A. $\hat{A}^\dagger \hat{B}^\dagger$
 - B. $\hat{B}^\dagger \hat{A}^\dagger$
 - C. $\hat{B}\hat{A}$
 - D. $(\hat{A}\hat{B})^{-1}$
5. What are the eigenvalues of the matrix $M = \begin{pmatrix} 3 & 2 \\ 2 & 0 \end{pmatrix}$? [2 marks]
- A. 1 and 2
 - B. 4 and -1
 - C. 3 and 0
 - D. 5 and -2

6. A system is in state $|\psi\rangle$ and observable A is measured, yielding eigenvalue a_n (which is non-degenerate). Immediately after the measurement, the state of the system is: [2 marks]

- A. $|\psi\rangle$ (unchanged)
- B. $\hat{A}|\psi\rangle$
- C. The eigenstate $|a_n\rangle$ corresponding to eigenvalue a_n
- D. A uniform superposition of all eigenstates of $\hat{A}$

7. A system is prepared in the energy eigenstate $|E_1\rangle$ at time $t = 0$. What is its state at time t ? [2 marks]

- A. $|E_1\rangle$
- B. $e^{+iE_1t/\hbar} |E_1\rangle$
- C. $e^{-iE_1t/\hbar} |E_1\rangle$
- D. $\cos(E_1t/\hbar) |E_1\rangle$

8. For a spin- $\frac{1}{2}$ particle, what are the eigenvalues of the operator $\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$? [2 marks]

- A. $\pm\hbar/2$
- B. $\hbar^2/4$
- C. $\frac{3}{4}\hbar^2$
- D. $\hbar^2$

9. Two identical fermions can each occupy one of two single-particle states $|\alpha\rangle$ or $|\beta\rangle$. How many distinct two-particle states are allowed? [2 marks]

- A. 1
- B. 2
- C. 3
- D. 4

10. What is the quark composition of a proton? [2 marks]

- A. udd
- B. uud
- C. uds
- D. $u\bar{d}$

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

Collisions, Impulse, and Centre of Mass

- (a) A 4 kg trolley moving at 6 m/s to the right collides head-on with a 2 kg trolley moving at 3 m/s to the left. The collision is perfectly inelastic (the trolleys stick together). Find the velocity of the combined mass after the collision, and state its direction of motion. 3
- (b) Calculate the total kinetic energy before and after the collision in part (a). What fraction of the initial kinetic energy is lost? 3
- (c) Suppose instead that the collision described in part (a) is *elastic*. Using conservation of momentum and kinetic energy (or the elastic collision formula provided on the formula sheet), find the velocity of each trolley after the collision. 4
- (d) A constant force acts on a 3 kg object for 0.5 s, changing its velocity from 2 m/s to 8 m/s (in the same direction). Calculate the impulse delivered to the object and the average force exerted. 2
- (e) Three point particles of masses 1 kg, 2 kg, and 3 kg are located at positions (0, 0), (3, 0), and (0, 4) metres, respectively. Find the position of the centre of mass of the system. 3

Question 2.

[15 marks]

Thermodynamic Cycle

Two moles of monatomic ideal gas ($\gamma = 5/3$) begin in state A at pressure $P_A = 1.00 \times 10^5$ Pa and temperature $T_A = 300$ K.

- (a) Find the volume V_A of the gas in state A. 3
- (b) The gas is heated at constant pressure (isobaric process) until the temperature reaches $T_B = 600$ K (state B). Find:
- (i) the volume V_B ;
 - (ii) the work done by the gas;
 - (iii) the heat absorbed by the gas.
- 4
- (c) From state B, the gas expands adiabatically until its temperature returns to 300 K (state C). Find the volume V_C and the pressure P_C . 4
- (d) The gas is then compressed isothermally at 300 K back to its original state A.
- (i) Calculate the work done on the gas in this step.
 - (ii) What is the total work done by the gas in one complete cycle?
 - (iii) Determine the efficiency of this cycle and compare it with the Carnot efficiency for the same temperature extremes.

4

Question 3.

[15 marks]

Coupled Oscillators and Pendulums

- (a) Two identical masses $m = 0.5$ kg are connected in a line by three identical springs, each with spring constant $k = 40$ N/m, in the arrangement:

wall — spring — mass 1 — spring — mass 2 — spring — wall

Let x_1 and x_2 denote the displacements of mass 1 and mass 2 from their respective equilibrium positions. Write down the equations of motion for the two masses. 3

- (b) Find the two normal mode angular frequencies. Describe the motion of the two masses in each normal mode. 4
- (c) At $t = 0$, mass 1 is displaced by 5 cm from equilibrium and mass 2 is at its equilibrium position. Both masses are released from rest. Express the subsequent motion of each mass as a superposition of the two normal modes. 3
- (d) A simple pendulum has length $L = 1.2$ m. Find its period of small oscillations. 2
- (e) A thin uniform rod of mass $M = 0.8$ kg and length $\ell = 0.6$ m is pivoted at one end and swings as a physical (compound) pendulum. Find:
- (i) the moment of inertia of the rod about the pivot;
 - (ii) the distance from the pivot to the centre of mass;
 - (iii) the period of small oscillations.

3

Question 4.

[15 marks]

Operators, Eigenvalues, and Expectation Values

Let $\{|1\rangle, |2\rangle\}$ be an orthonormal basis. Consider the operator

$$\hat{A} = 2|1\rangle\langle 1| - |2\rangle\langle 2| + 2|1\rangle\langle 2| + 2|2\rangle\langle 1|.$$

- (a) Show that $\hat{A}$ is Hermitian. Write $\hat{A}$ as a 2×2 matrix in the $\{|1\rangle, |2\rangle\}$ basis. 3
- (b) Find the eigenvalues and normalised eigenvectors of $\hat{A}$. 4
- (c) A system is prepared in the state

$$|\psi\rangle = \frac{3}{5}|1\rangle + \frac{4}{5}|2\rangle.$$

If observable A is measured, what are the possible outcomes and their probabilities? Calculate the expectation value $\langle A \rangle$. 4

- (d) Suppose the measurement in part (c) yields the *larger* eigenvalue. The system then evolves under the Hamiltonian

$$\hat{H} = E_0|1\rangle\langle 1| + 2E_0|2\rangle\langle 2|,$$

where $E_0 > 0$. Find the state of the system at time t , and determine the probability of again measuring the larger eigenvalue of A at that time. 4

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v'_1 = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m